

Shiva Rami Reddy Madira

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PROFESSIONAL SUMMARY

DevOps Engineer with **4+ years** of experience designing and building cloud infrastructure across **AWS, Azure, and GCP**. At Citibank, I own the multi cloud **Kubernetes** infrastructure for internal payment processing systems, cutting incident response time from 60+ minutes to under 40 minutes. Earlier work includes EKS based healthcare infrastructure at TCS, increasing deployment frequency from bi-weekly to multiple per week and a full Azure cloud migration at Wipro, reducing provisioning time by 65%.

EXPERIENCE

Cloud Platform Engineer

June 2025 – Present

Citibank

San Jose, CA

- Build and operate a secure **multi cloud Kubernetes platform** on **AWS and GCP**, modernizing internal payment processing systems to support high volume financial workloads at Citibank.
- Design cloud landing zones with network segmentation, IAM boundaries, and environment isolation to ensure secure access across development, staging, and production environments.
- Develop reusable **Terraform modules** to standardize infrastructure provisioning across multiple cloud accounts, reducing configuration inconsistencies by **40%**.
- Implement GitOps based deployment workflows using **ArgoCD**, enabling engineering teams to deploy application changes through version controlled manifests.
- Manage **EKS and GKE clusters**, configuring namespace isolation, autoscaling, and network policies to support high-volume internal payment services.
- Deploy blue/green and canary deployment strategies, reducing service disruptions by **25%** during critical payment system updates.
- Integrate static code analysis and container image scanning into CI/CD pipelines, preventing vulnerable builds from reaching production.
- Establish centralized monitoring with Prometheus and cloud native logging tools, defining **SLOs** and alerting standards that reduced incident response time from over an hour to under 40 minutes.
- Onboard EKS clusters onto **Dynatrace**, using **Davis AI** for automated anomaly detection and root cause analysis, reducing false positive alerts by **30%** and improving incident response time in production.
- Configure **HashiCorp Vault** for secrets management, enforcing RBAC and periodic rotation policies to maintain least privilege access across environments.

Systems Engineer

June 2022 – July 2023

Tata Consultancy Services

Hyderabad

- Architected secure AWS infrastructure using modular Terraform for **VPCs, multi-AZ RDS, Auto Scaling Groups, ALBs, EC2, IAM, and S3**, reducing time to deploy new environments by **55%**.
- Standardized Terraform S3 remote state management with **DynamoDB state locking**, eliminating configuration conflicts and improving deployment reliability across the team.
- Built and managed **CI/CD pipelines with Jenkins and GitLab CI** to deliver updates from bi-weekly releases to multiple deployments per week.
- Migrated containerized healthcare applications into Amazon EKS with **Helm** charts and ArgoCD GitOps workflows, configuring cluster wide **RBAC** and autoscaling policies to meet **HIPAA adjacent compliance and SLA**.
- Tuned Kubernetes workloads by analyzing scheduler events and node resource utilization, resolving node pressure and PVC binding issues to restore optimal application performance.
- Configured readiness/liveness probes, resource limits, and autoscaling policies in EKS, improving application resilience and reducing crash restarts by **20%** during traffic spikes.
- Resolved production incidents in EKS environments through root cause analysis, implementing corrective changes that maintained **99.9%** application availability.
- Lowered Mean Time to Recovery (**MTTR**) by **30%** through centralized observability using **Prometheus, Grafana, CloudWatch and ELK Stack** for log aggregation, alerting, and distributed tracing across production.
- Enabled API activity monitoring and audit logging with **CloudTrail** for compliance and incident analysis.
- Reduced monthly AWS infrastructure costs by **15%** through S3 lifecycle policies, Auto Scaling tuning, and removal of unused EBS volumes.

- Integrated **Snowflake** with AWS S3 for secure data pipeline workflows, configuring virtual warehouses and RBAC policies to support analytics teams while maintaining data access controls.
- Diagnosed and resolved a critical production outage caused by **IAM misconfiguration**, restoring full application availability and implementing stronger least-privilege policies to prevent recurrence.

Project Engineer

Sept 2020 – May 2022

Wipro

Bengaluru

- Migrated **30+** legacy and AWS hosted workloads to Azure, standardizing infrastructure and enabling hybrid cloud scalability across the organization.
- Automated provisioning of Azure infrastructure with Terraform and ARM templates, compressing environment setup from multiple days to under 6 hours, a **65%** reduction in provisioning time.
- Combined Terraform with **Ansible** to automate the post-provisioning configuration of Azure VMs, including package installation and service setup, resulting in consistent server builds across environments.
- Designed multi AZ deployments with public/private subnet separation, load balancing, and autoscaling for high availability business critical applications.
- Established **Azure DevOps CI/CD pipelines** for container builds and AKS deployments, reducing manual release efforts by **50%**.
- Containerized and deployed applications to AKS, configuring readiness/liveness probes, Horizontal Pod Autoscalers, and Pod Disruption Budgets to improve workload stability.
- Managed Kubernetes deployments declaratively through ArgoCD GitOps workflows, reducing configuration drift across environments by **35%**.
- Eliminated frequent memory configuration issues in AKS workloads, reducing production restart frequency by **25%** during peak traffic periods.
- Integrated **SonarQube** quality gates and container vulnerability scanning into CI/CD pipelines, enforcing **DevSecOps** standards across all application deployments.
- Standardized AKS node pool configurations and VM size policies to prevent resource over utilization and improve workload efficiency across migrated applications.
- Managed **Azure SQL databases** across migrated workloads, configuring performance monitoring, backup policies, and access controls to maintain availability and data integrity.

TECHNICAL SKILLS

Cloud Platforms: AWS (EC2, VPC, RDS, S3, IAM, ECS, EKS, Lambda), Microsoft Azure (VMs, VNETs, NSGs, AKS, Azure DevOps, Key Vault, Monitor), GCP (GKE, IAM, VPC)

Infrastructure as Code: Terraform, ARM Templates, Azure Bicep, CloudFormation, Terragrunt

Containerization & Orchestration: Docker, Kubernetes, Helm

CI/CD Tools: Github Actions, Jenkins, GitLab CI, Azure DevOps Pipelines, ArgoCD, Kustomize

Version Control Tools: Git, Bitbucket, GitHub

Configuration Management Tools: Ansible, Ansible Tower

Observability & Monitoring: Prometheus, Grafana, Datadog, ELK Stack, OpenTelemetry, CloudWatch, Azure Monitor, Log Analytics, Dynatrace

DevSecOps: SonarQube, Trivy, Container Image Scanning, HashiCorp Vault

SRE Practices: SLO/SLI definition, Error Budgets, Blameless Postmortems, Incident Response, Capacity Planning, MTTR/MTTD Optimization

Databases & Data Platforms: PostgreSQL (Amazon RDS), Azure SQL Database, Snowflake, SQL

Networking & Load Balancing: NGINX, Route 53, ELB/ALB, VPC, Subnets, Security Groups

Scripting: Python, Bash, PowerShell, YAML, JSON

Operating Systems: Linux, Windows Server

CERTIFICATIONS

- Microsoft Certified: Azure Developer Associate
- Microsoft Certified: Azure Fundamentals

EDUCATION

Texas Tech University

Lubbock, TX

Masters of Computer Science

Aug 2023 – May 2025

Sathyabama Institute of Science and Technology

Chennai

Bachelor of Engineering in Computer Science and Engineering

Aug 2016 – May 2020